

Bitwise differential trail through **LBlock** found by SAT

CiVerLy

July 24, 2026

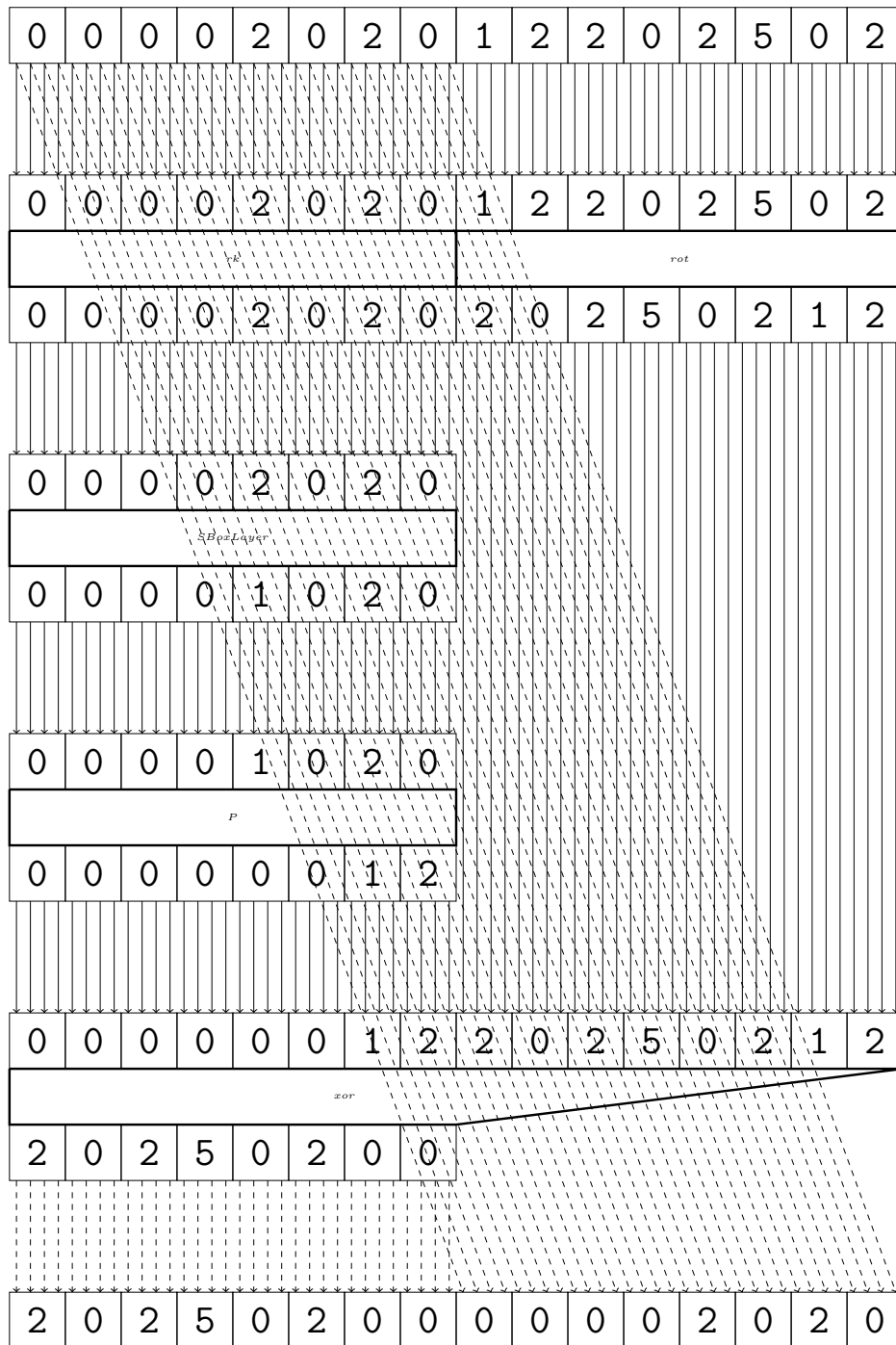
1 LBlock

Maximal differential probability: 2^{-44}

0	0	0	0	2	0	2	0	1	2	2	0	2	5	0	2
0	0	0	0	2	0	2	0	1	2	2	0	2	5	0	2
LBlock round															
2	0	2	5	0	2	0	0	0	0	0	2	0	2	0	
2	0	2	5	0	2	0	0	0	0	0	2	0	2	0	
LBlock round															
0	2	0	1	0	0	0	0	2	0	2	5	0	2	0	0
0	2	0	1	0	0	0	0	2	0	2	5	0	2	0	0
LBlock round															
0	0	0	2	0	0	2	0	0	2	0	1	0	0	0	0
0	0	0	2	0	0	2	0	0	2	0	1	0	0	0	0
LBlock round															
0	0	0	0	0	0	0	0	0	0	2	0	0	2	0	
0	0	0	0	0	0	0	0	0	0	2	0	0	2	0	
LBlock round															
0	2	0	0	2	0	0	0	0	0	0	0	0	0	0	0
0	2	0	0	2	0	0	0	0	0	0	0	0	0	0	0
LBlock round															
2	0	0	0	0	0	1	0	0	2	0	0	2	0	0	0
2	0	0	0	0	0	1	0	0	2	0	0	2	0	0	0
LBlock round															
0	0	0	0	0	0	0	0	2	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0	2	0	0	0	0	0	1	0
LBlock round															
0	0	0	0	1	0	2	0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	2	0	0	0	0	0	0	0	0	0
LBlock round															
0	0	0	0	0	0	1	2	0	0	0	0	1	0	2	0
0	0	0	0	0	0	1	2	0	0	0	0	1	0	2	0
LBlock round															
0	0	1	0	2	1	0	2	0	0	0	0	0	1	2	
0	0	1	0	2	1	0	2	0	0	0	0	0	1	2	
LBlock round															
0	0	0	1	3	3	1	0	0	0	1	0	2	1	0	2
0	0	0	1	3	3	1	0	0	0	1	0	2	1	0	2
Final step															
0	0	1	0	2	1	0	2	0	0	0	1	3	3	1	0
0	0	1	0	2	1	0	2	0	0	0	1	3	3	1	0

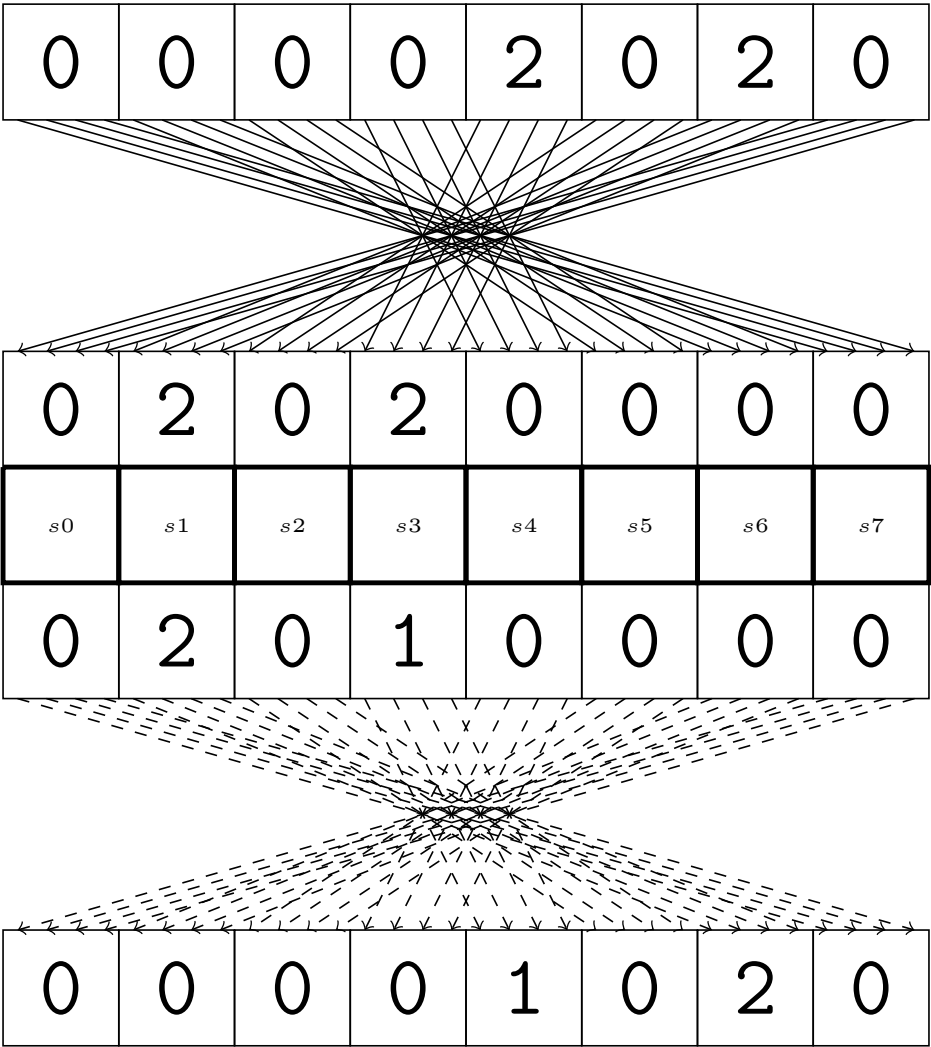
2 LBlock_round

Maximal differential probability: $2^{-4.0}$



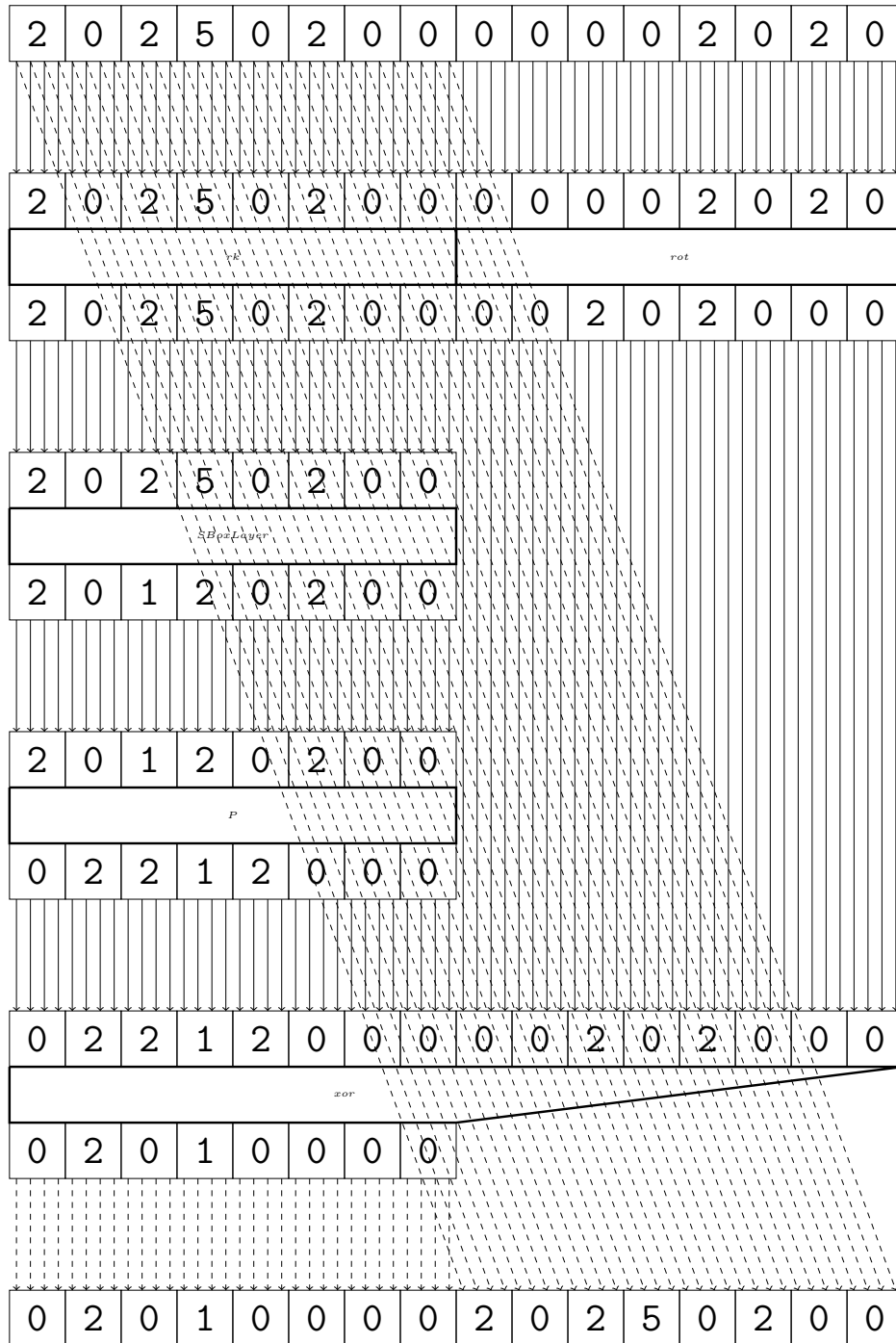
3 SBoxLayer

Maximal differential probability: $2^{-4.0}$



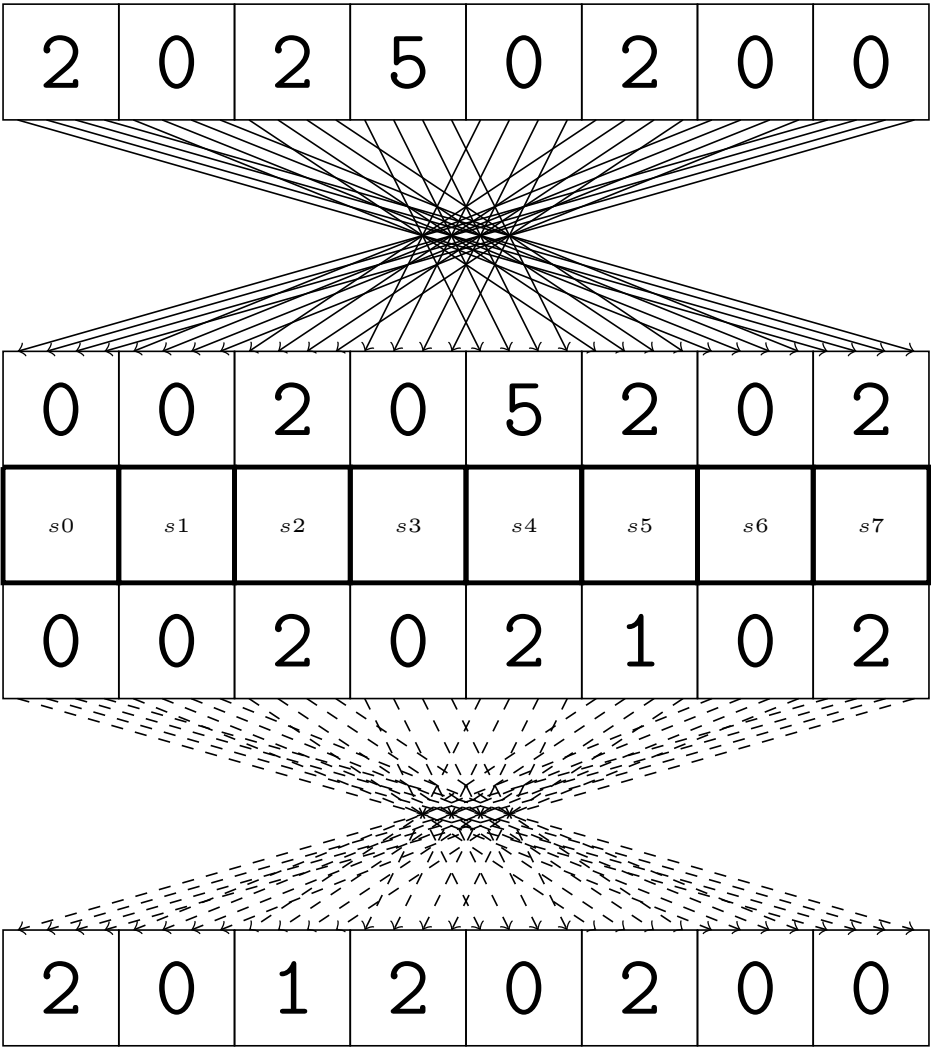
4 LBlock_round

Maximal differential probability: $2^{-8.0}$



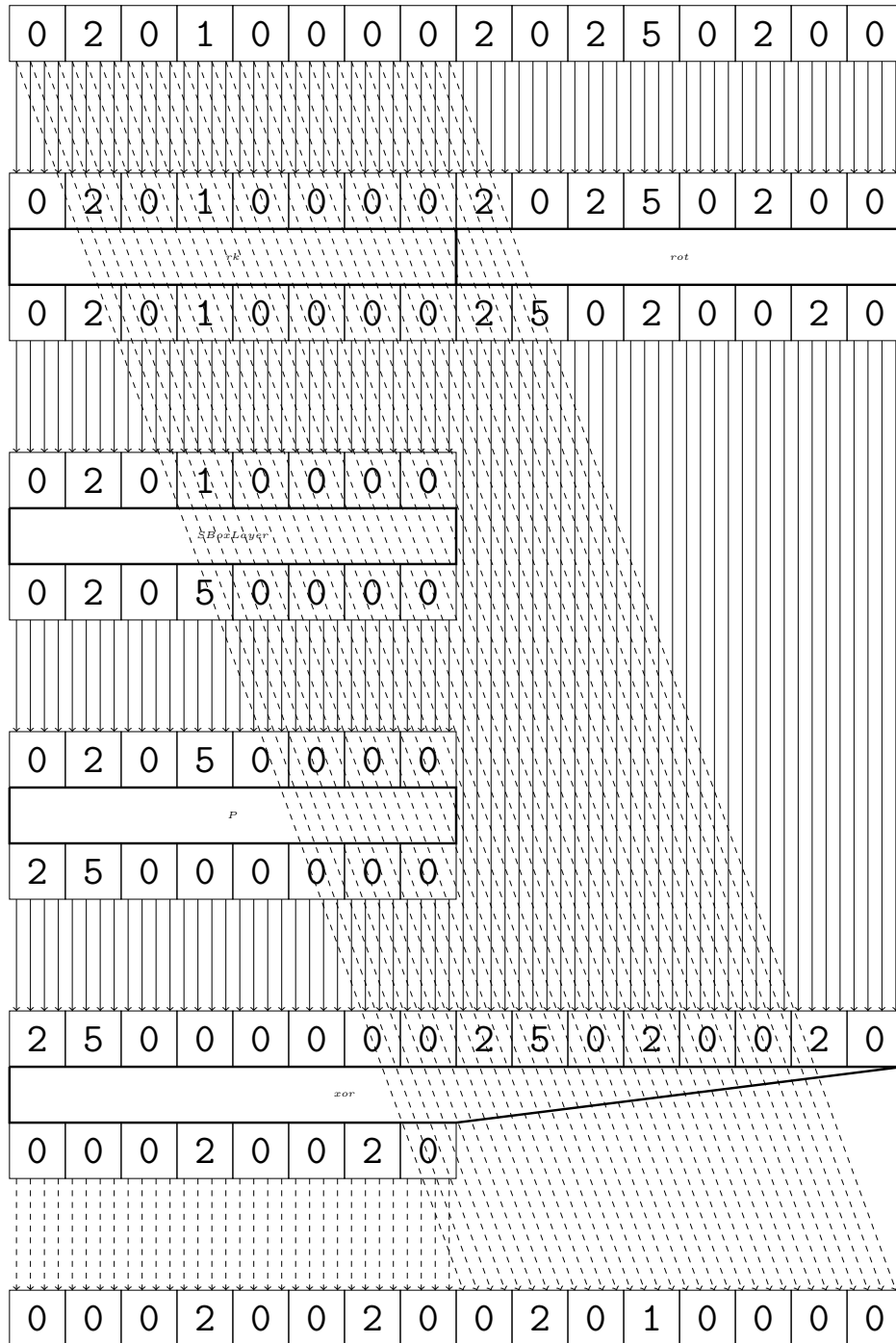
5 SBoxLayer

Maximal differential probability: $2^{-8.0}$



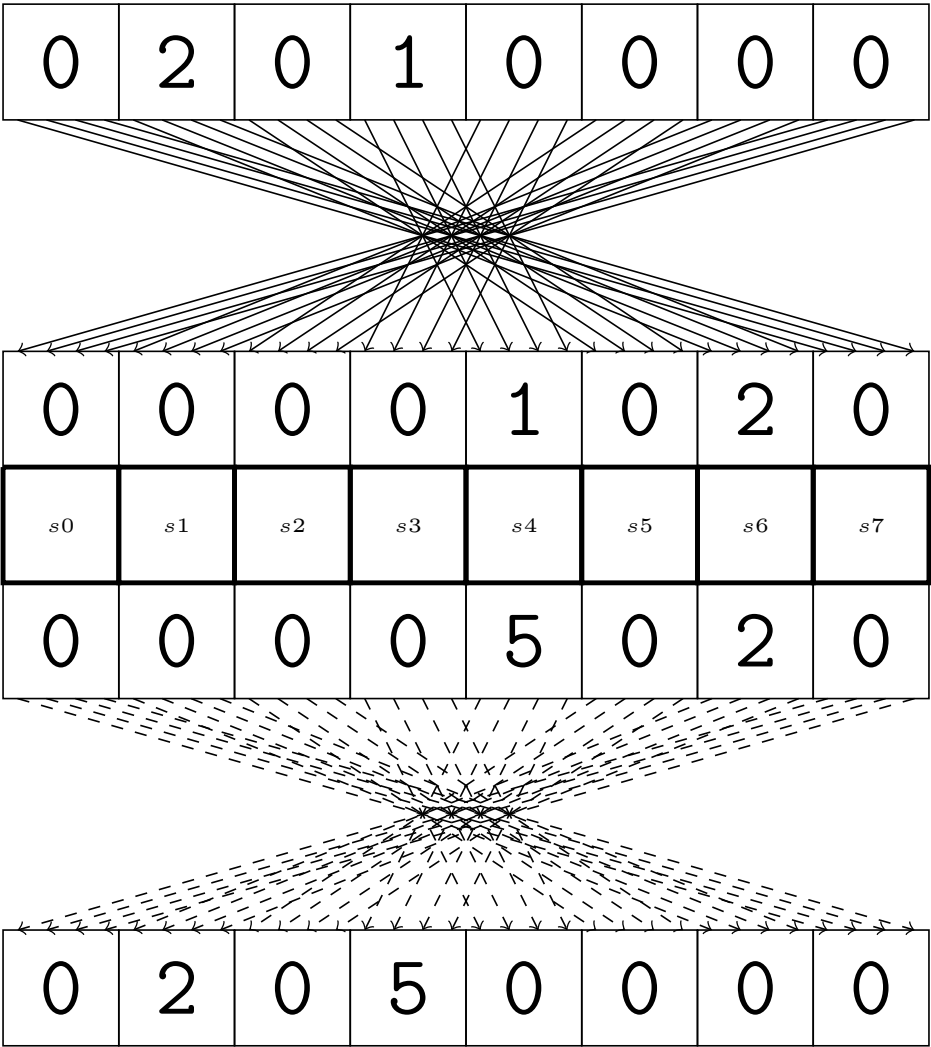
6 LBlock_round

Maximal differential probability: $2^{-4.0}$



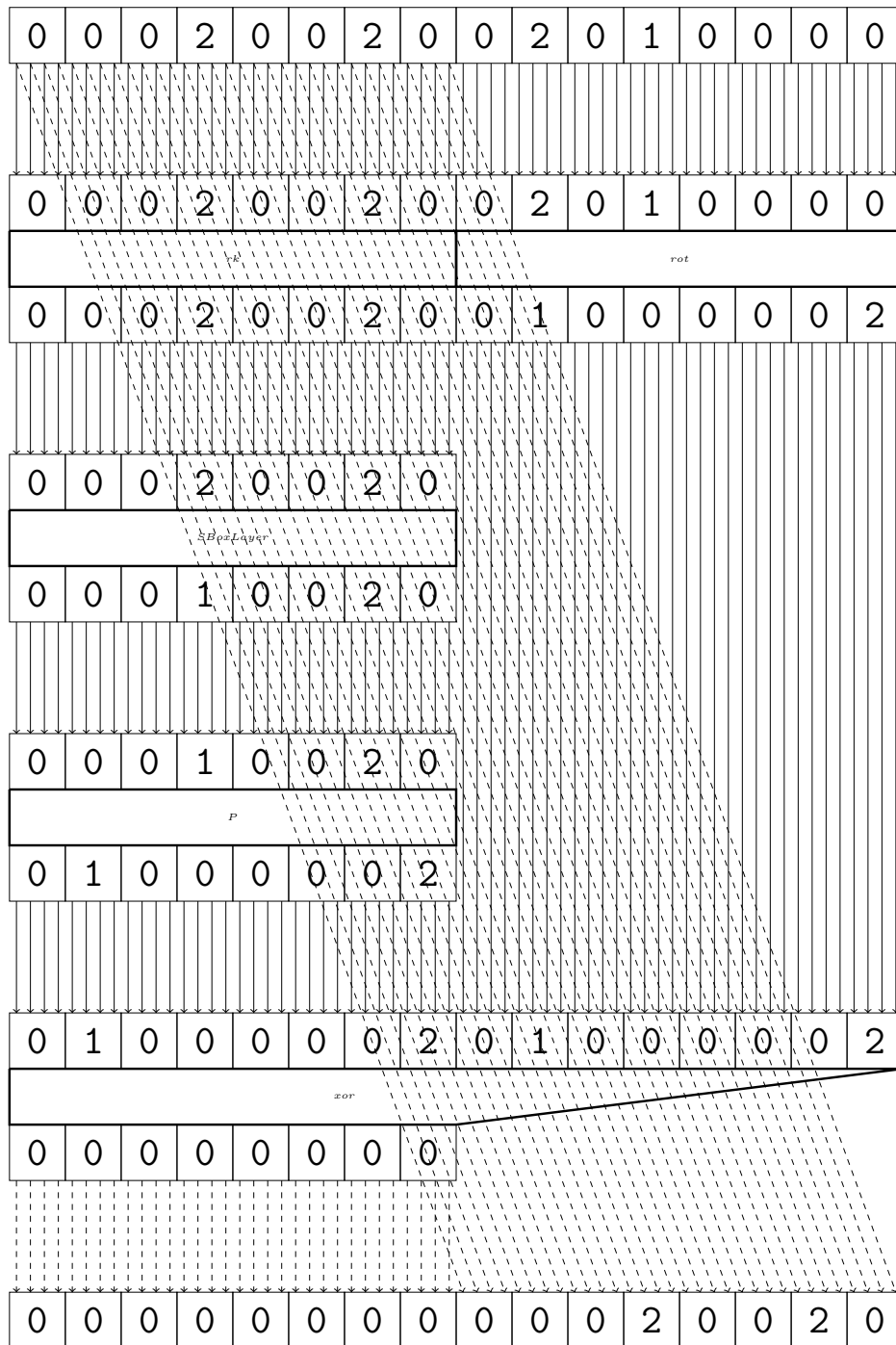
7 SBoxLayer

Maximal differential probability: $2^{-4.0}$



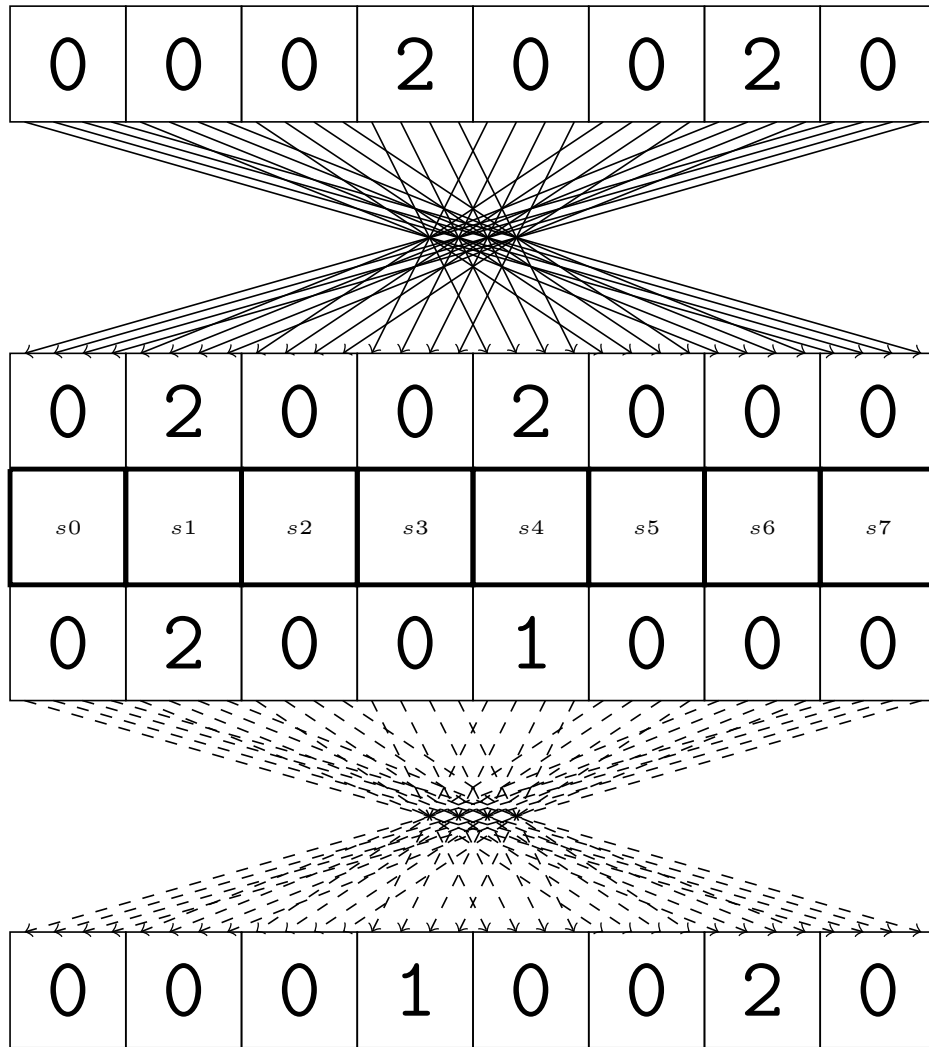
8 LBlock_round

Maximal differential probability: $2^{-4.0}$



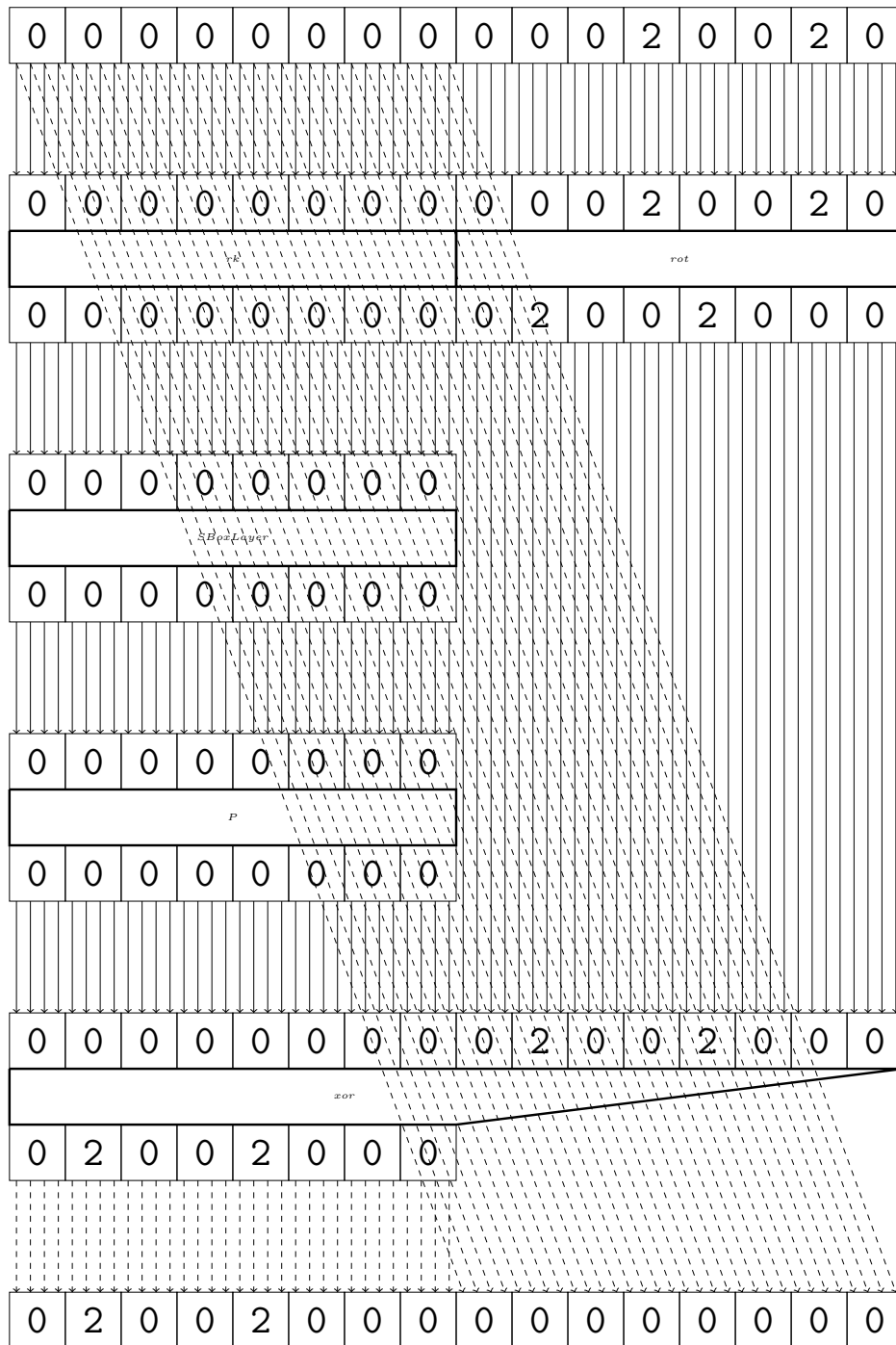
9 SBoxLayer

Maximal differential probability: $2^{-4.0}$



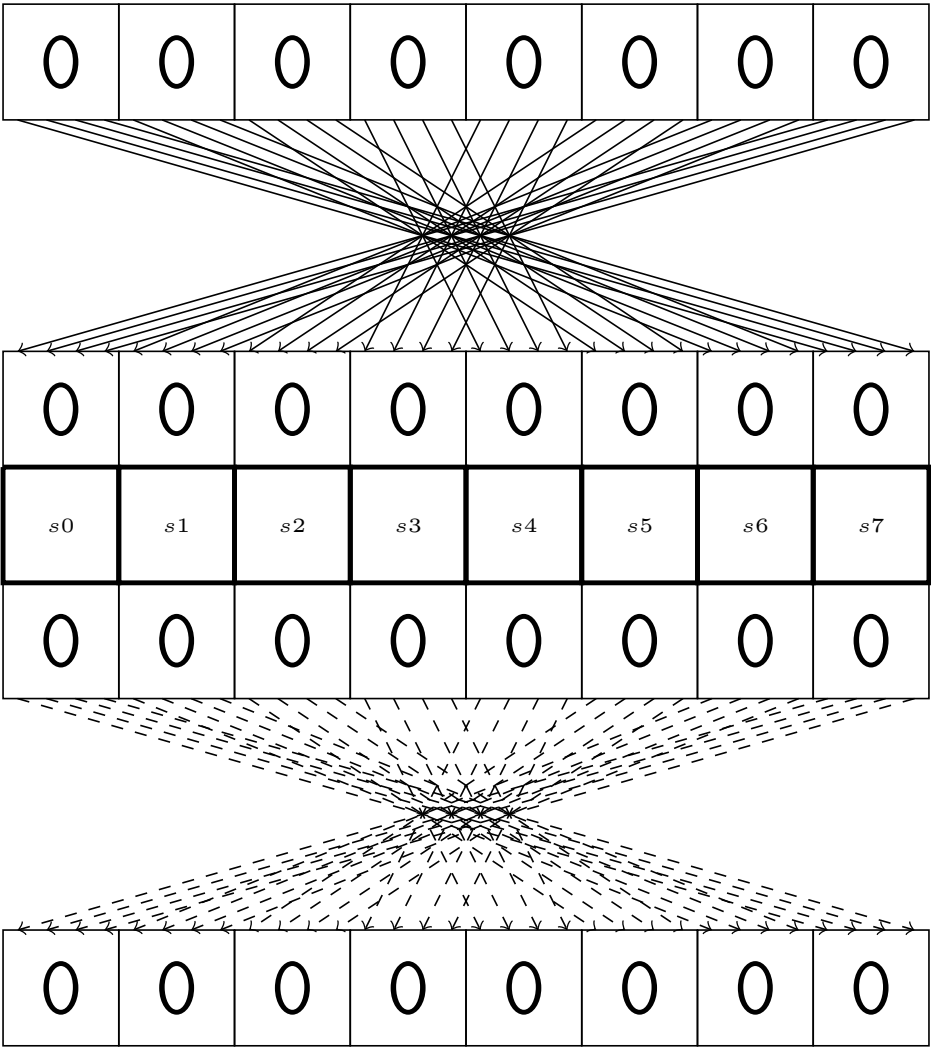
10 LBlock_round

Maximal differential probability: $2^{-0.0}$



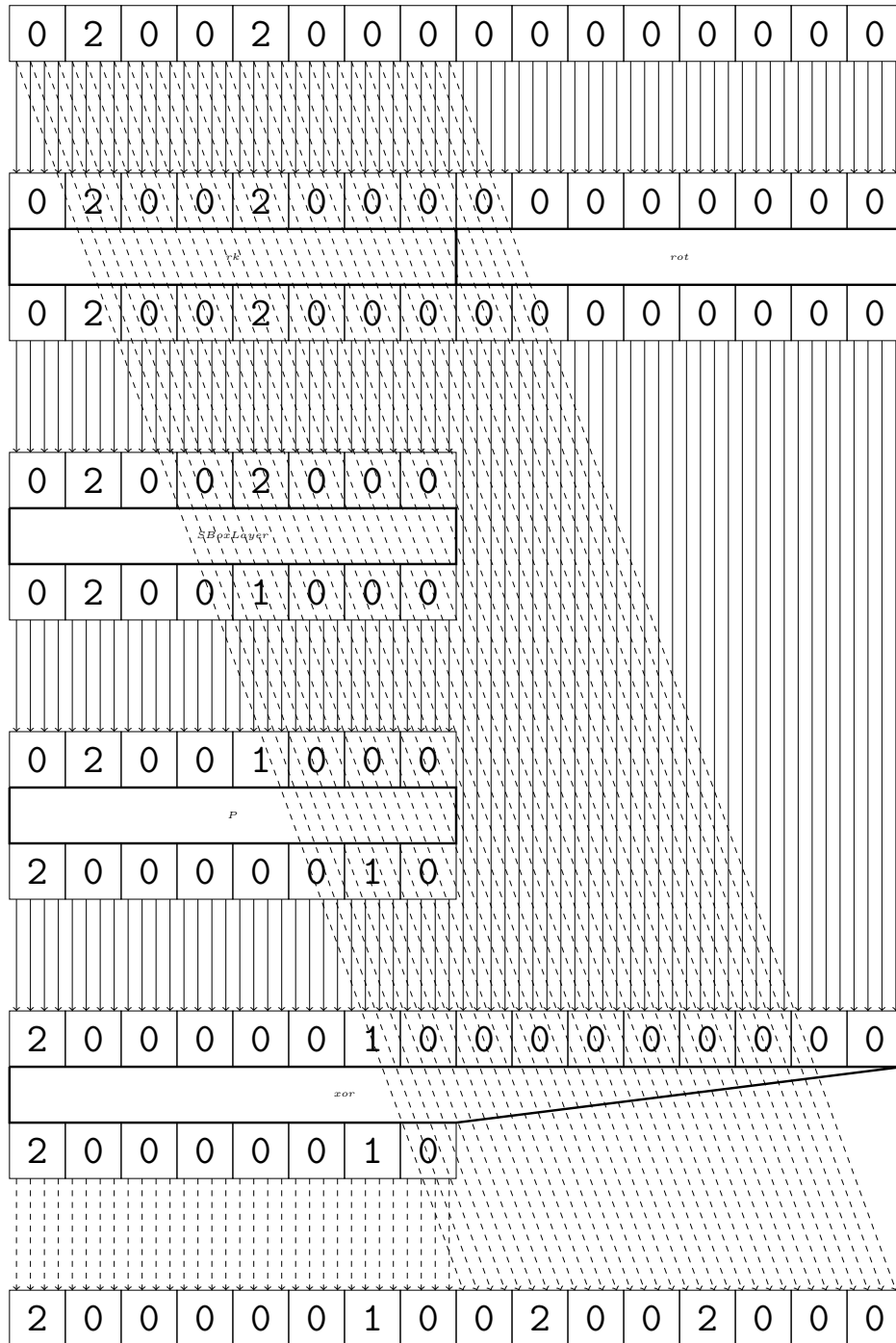
11 SBoxLayer

Maximal differential probability: $2^{-0.0}$



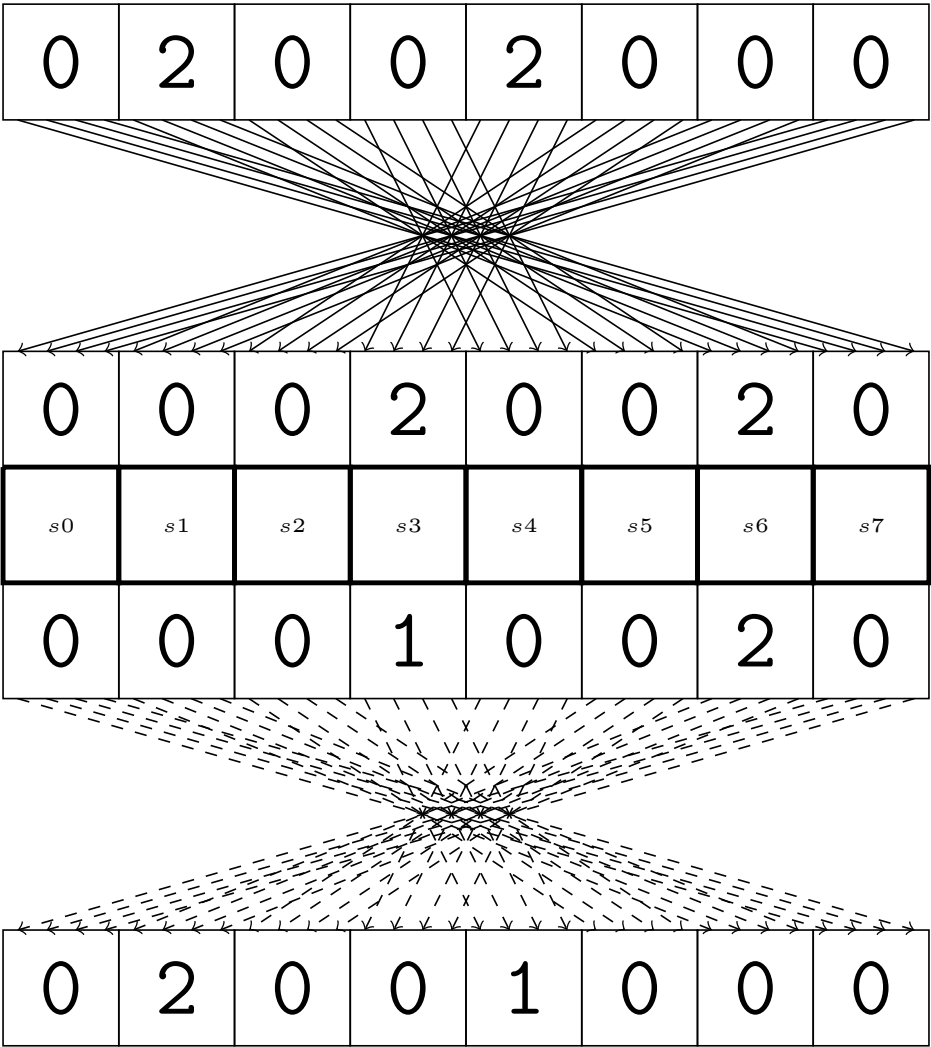
12 LBlock_round

Maximal differential probability: $2^{-4.0}$



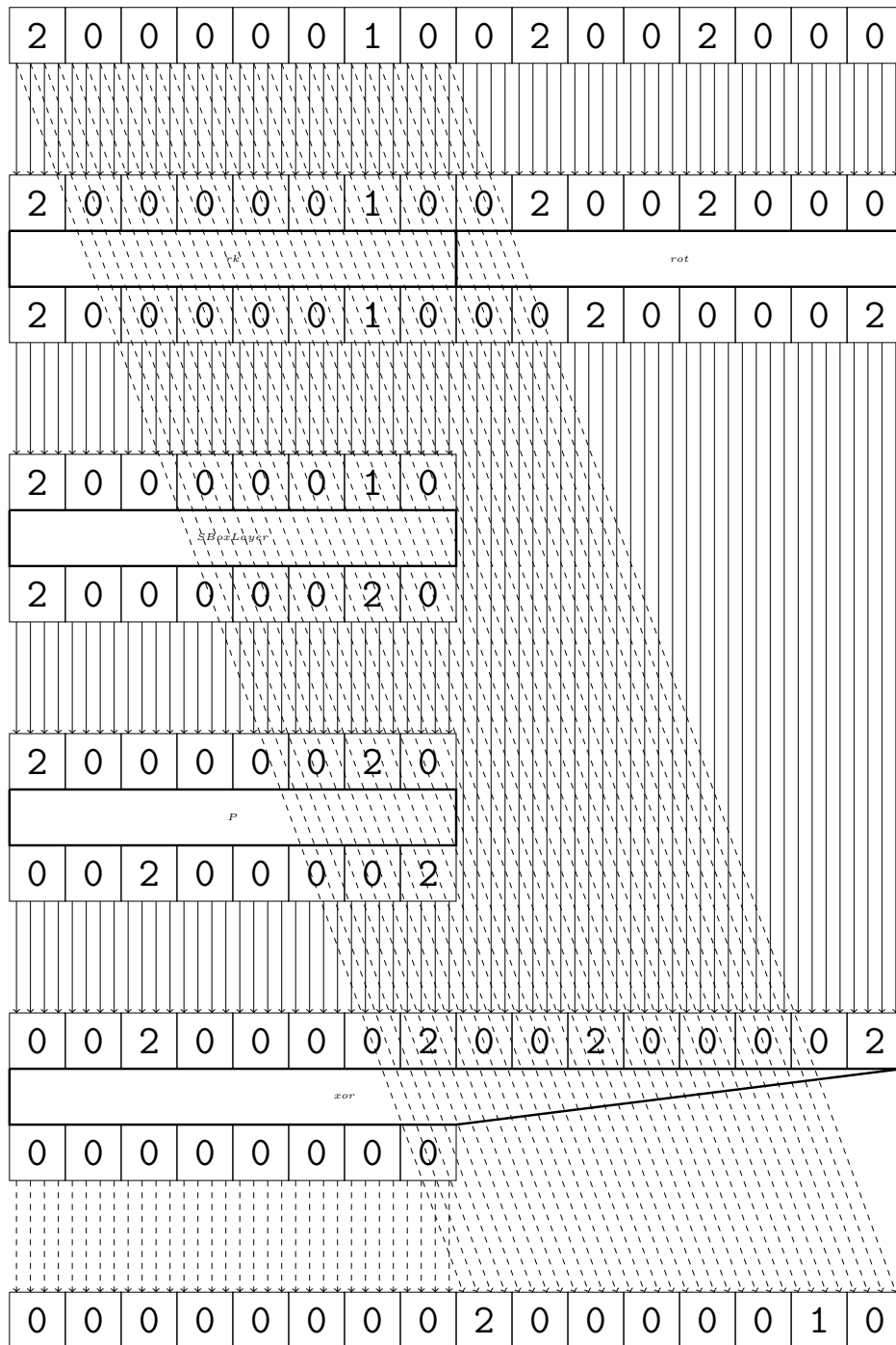
13 SBoxLayer

Maximal differential probability: $2^{-4.0}$



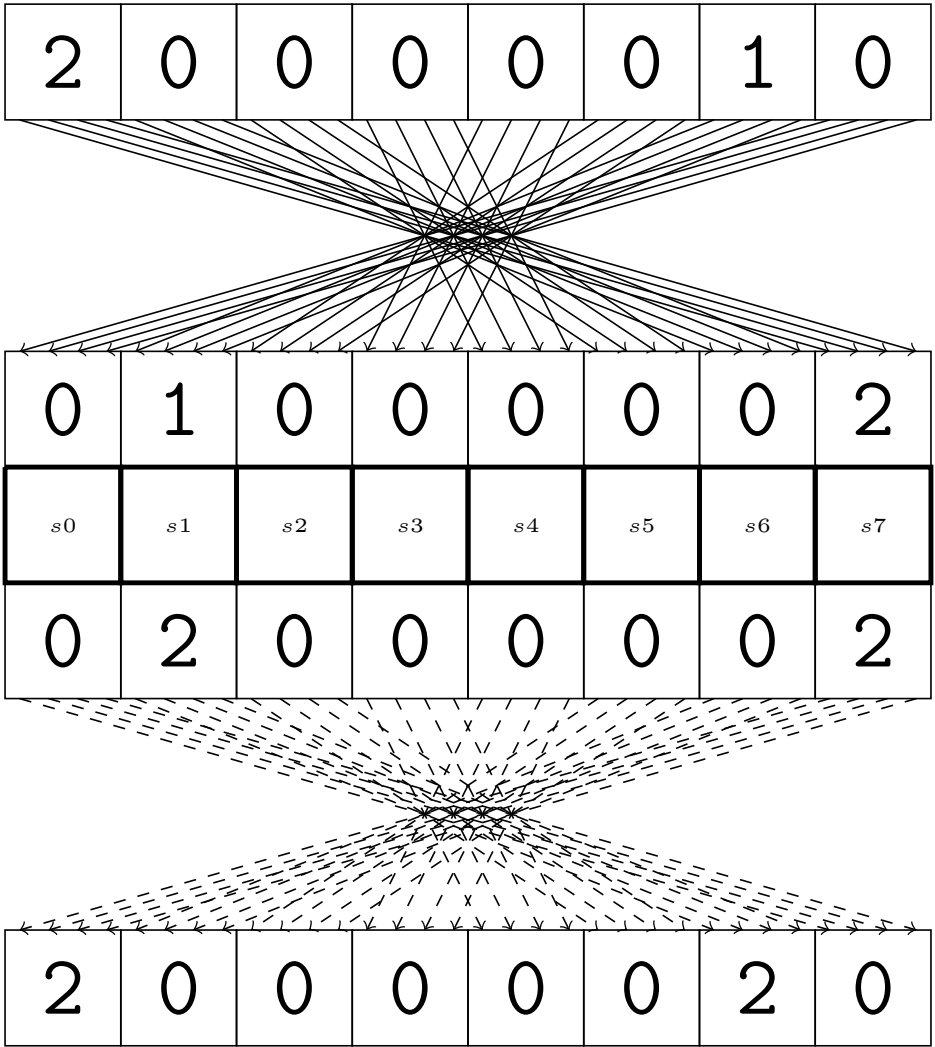
14 LBlock_round

Maximal differential probability: $2^{-4.0}$



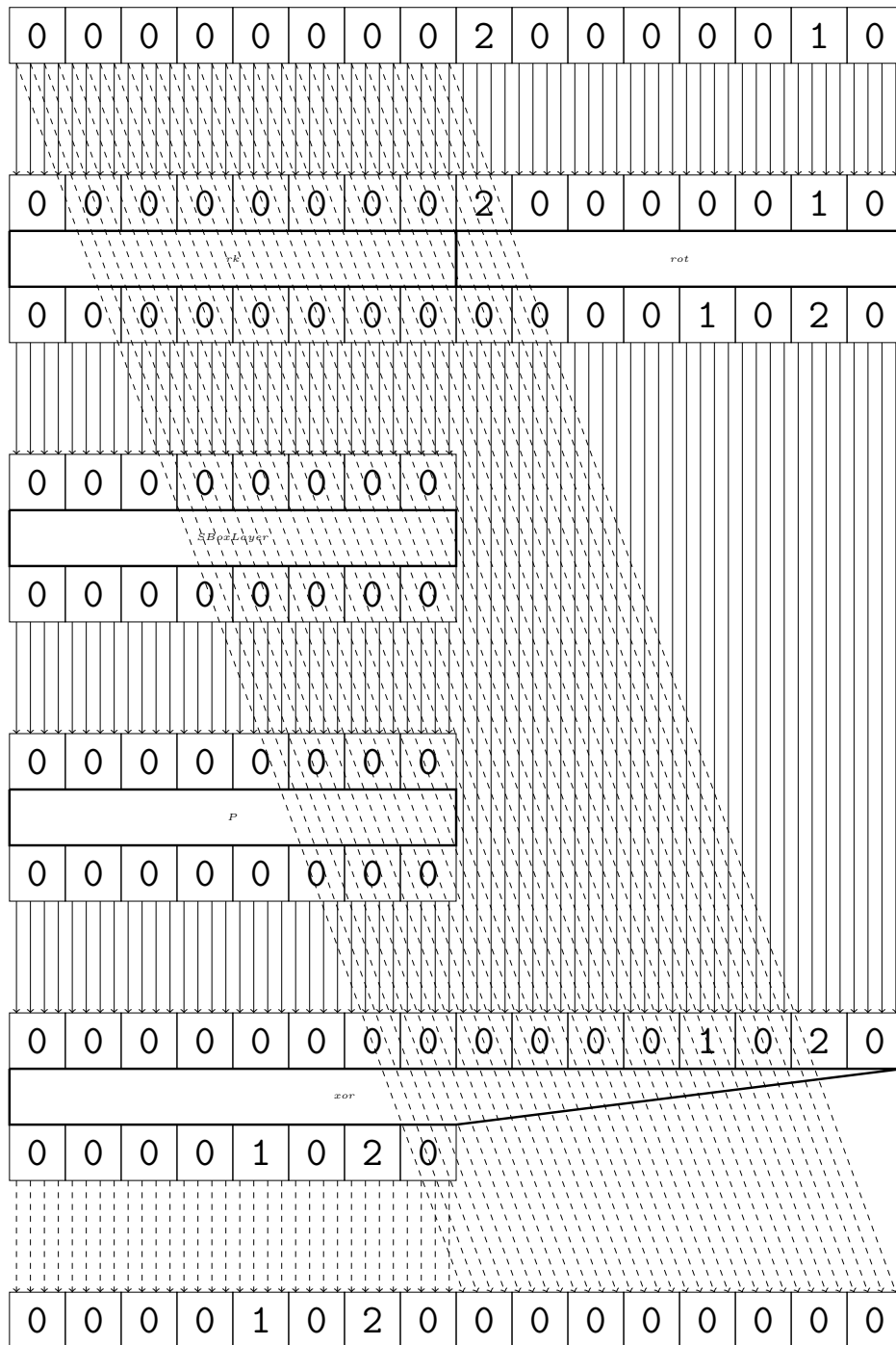
15 SBoxLayer

Maximal differential probability: $2^{-4.0}$



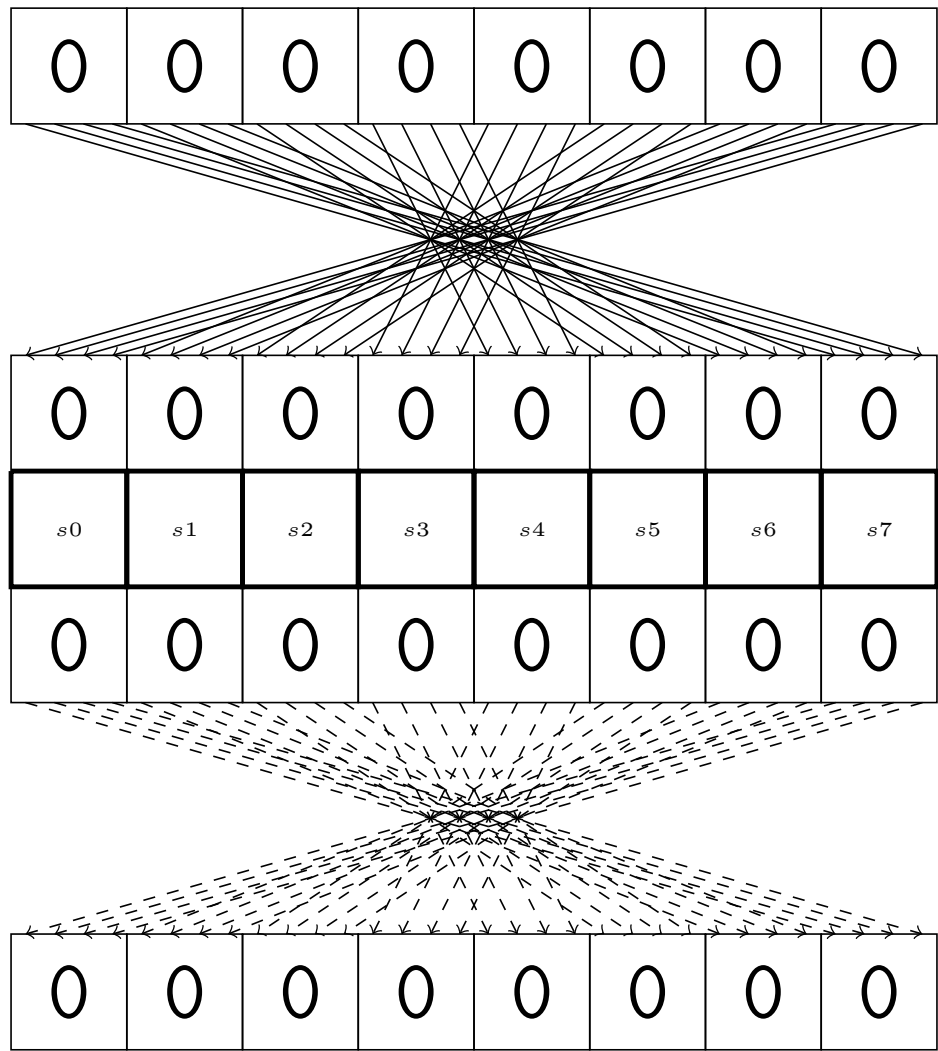
16 LBlock_round

Maximal differential probability: $2^{-0.0}$



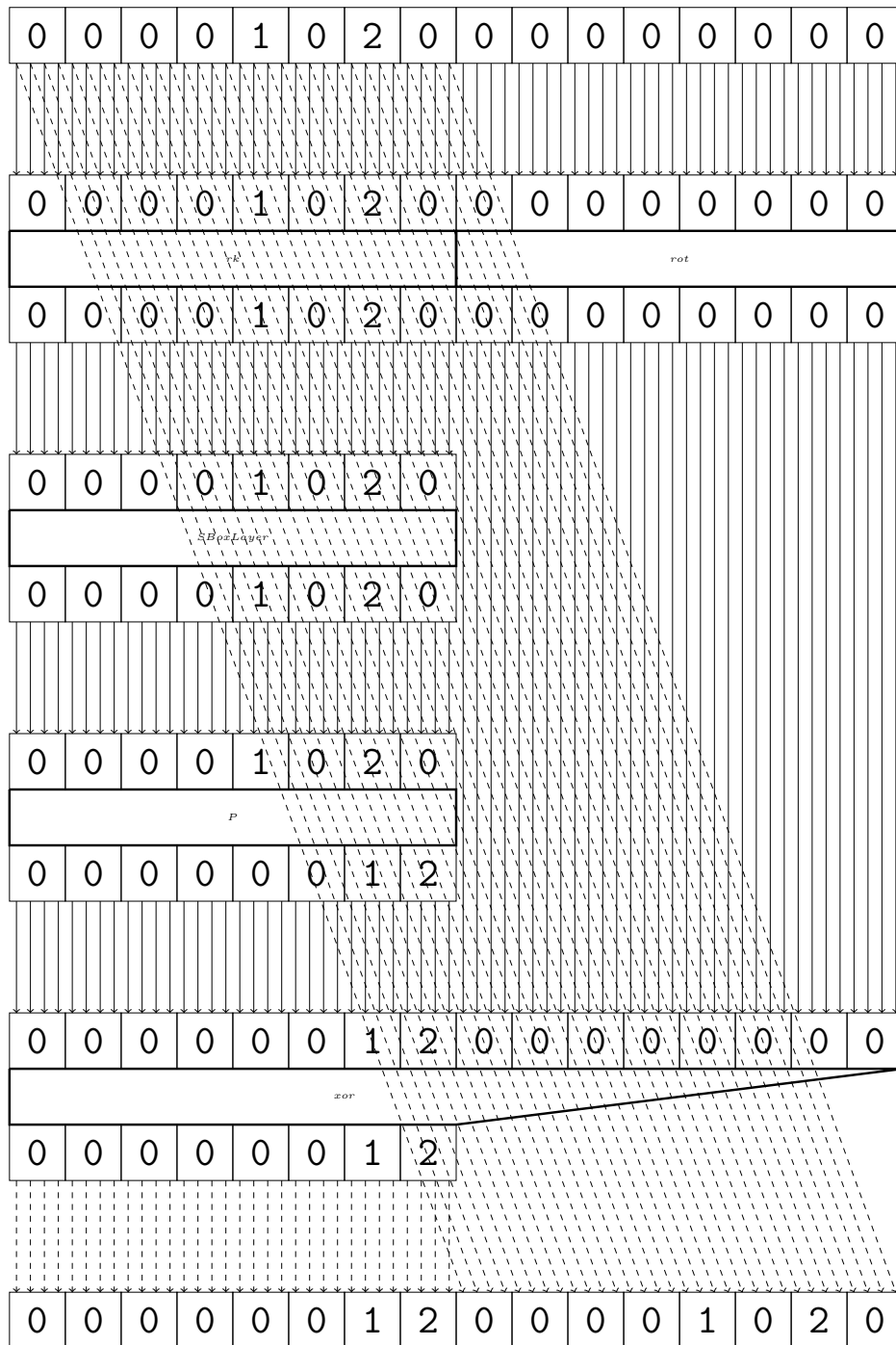
17 SBoxLayer

Maximal differential probability: $2^{-0.0}$



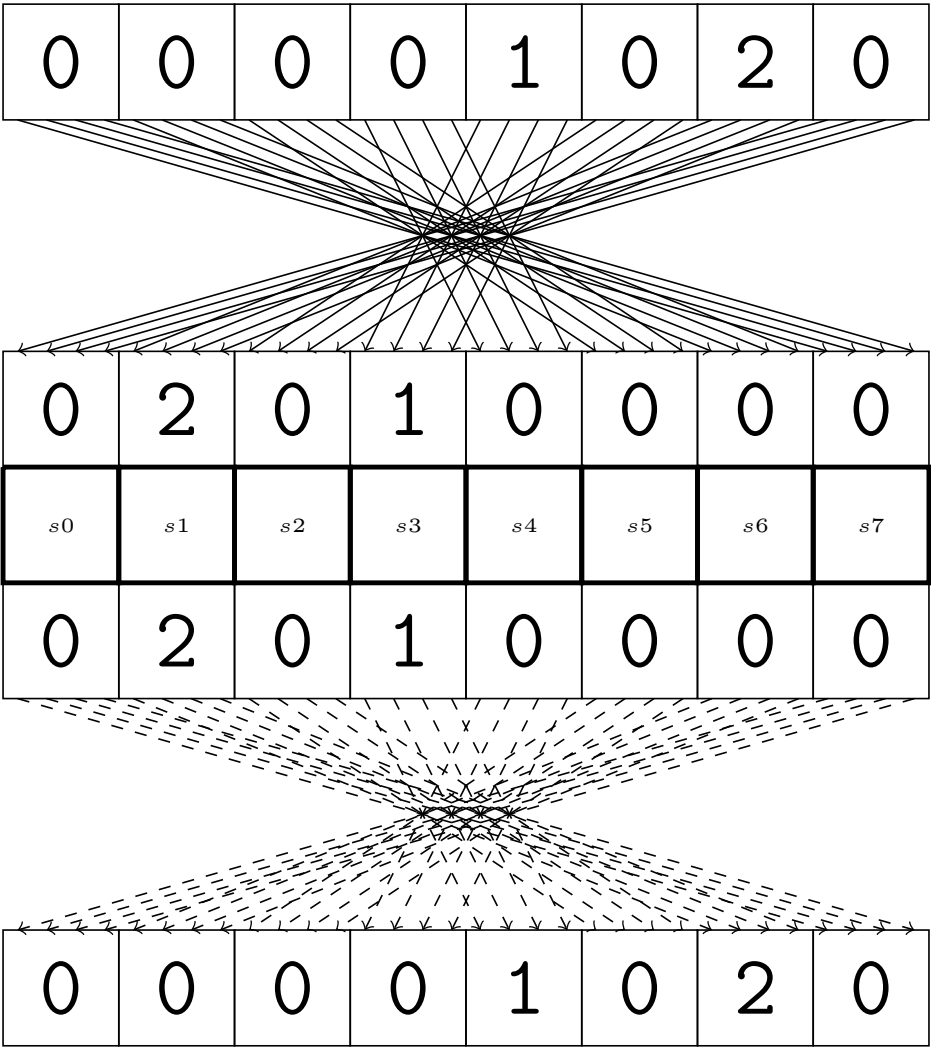
18 LBlock_round

Maximal differential probability: $2^{-4.0}$

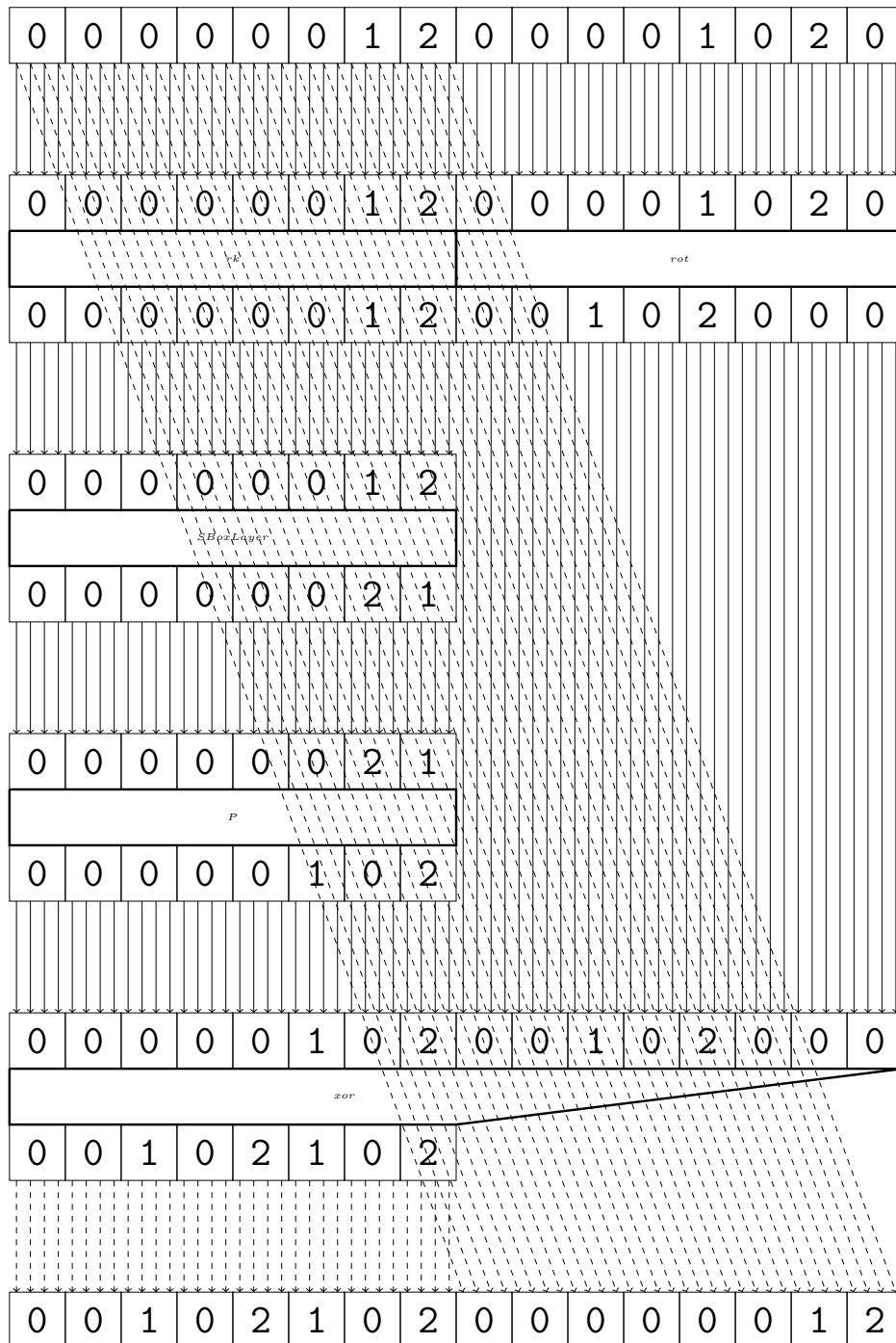


19 SBoxLayer

Maximal differential probability: $2^{-4.0}$

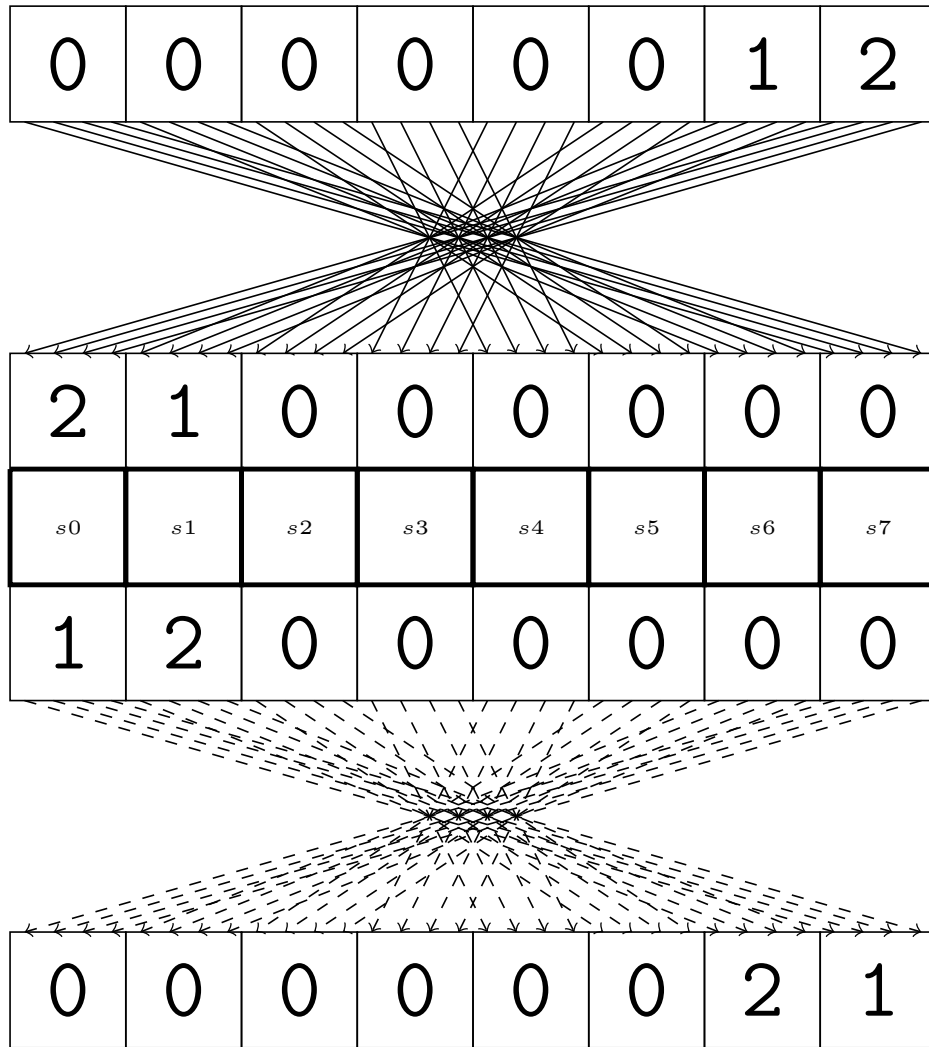


20 LBlock_round

Maximal differential probability: $2^{-4.0}$ 

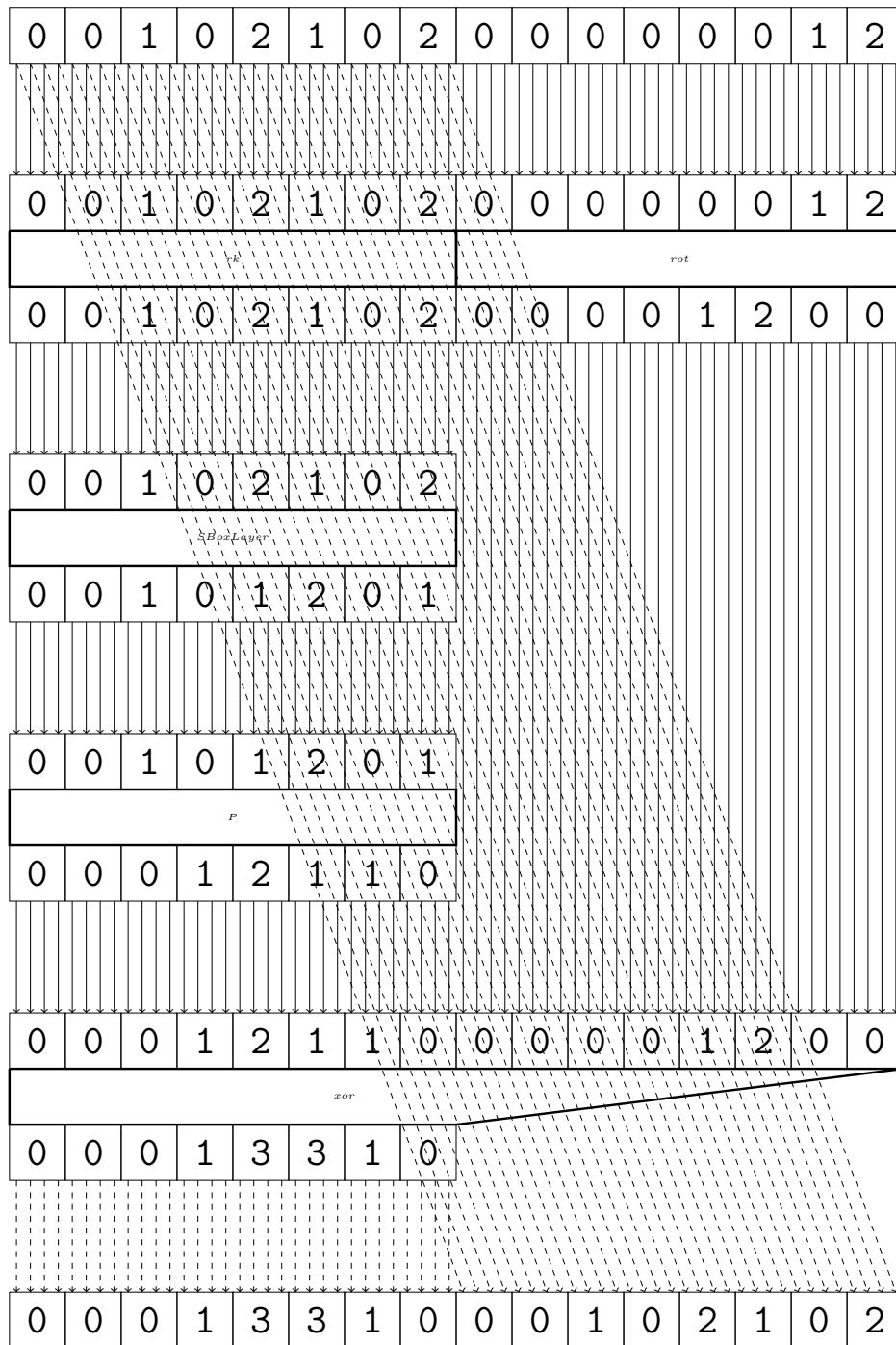
21 SBoxLayer

Maximal differential probability: $2^{-4.0}$



22 LBlock_round

Maximal differential probability: $2^{-8.0}$



23 SBoxLayer

Maximal differential probability: $2^{-8.0}$

